RailConnect 2.0: Autonomous Multi-WAN Failover & Train LAN QoS Orchestration

Technical Evaluation Report & Collaborative Testing Data Request Proposal

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Target Domain: High-Speed Passenger & Freight Rail (160–350 km/h)

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Status: Ready for Testing Collaboration / Pilot Request

1. Executive Summary

Modern high-speed passenger rail networks demand unbroken connectivity for two divergent requirements: **critical mission-operational systems** (ETCS train signaling, telemetry, driver VoIP intercom, CCTV monitoring) and **high-density passenger services** (guest Wi-Fi, streaming, web access). At cruising velocities between 160 km/h and 350 km/h, single-operator cellular SIM links frequently suffer severe Doppler shift frequency degradation, rapid cell tower handovers every 15–25 seconds, physical cuttings/tunnels, and carriage Faraday cage attenuation.

RailConnect 2.0 resolves this through an edge-intelligent architecture that binds a heterogeneous 4-link uplink matrix (Dual 5G Cellular carriers, Starlink LEO Satellite, and Gigabit Trackside Wi-Fi 6E) with an autonomous Machine Learning Predictive Failover Engine and dynamic **Train LAN QoS Traffic Slicing**. Rather than waiting for a connection drop to trigger a reactive failover, RailConnect evaluates real-time RF attenuation rates and geofenced track hazards to proactively execute seamless handovers within 30–50 ms. This document presents the technical architecture, dynamic LAN adaptation mechanisms, simulation results, and the data collaboration proposal for physical train modem logs.

2. Physics of High-Speed Rail Telecommunications

Wireless propagation along electrified railway corridors faces severe physical constraints:

- **Doppler Frequency Shift & Demodulation Loss:** At 250+ km/h in 3.5 GHz (5G C-Band), carrier frequency offsets reach several hundred Hertz, degrading signal-to-noise ratios (SINR) and elevating block error rates (BLER).
- **Rapid Handoff Oscillation:** Trains travel through cell tower radii within 15–30 seconds. In standard routing architectures, this causes packet storms, route flap, and DNS session dropouts.
- **Coach Body Penetration Loss:** Stainless steel and carbon-composite carriage shells cause 18–25 dB attenuation, isolating onboard passengers from trackside macro cells.
- **Topographical Blackouts:** Mountain gorges, deep cuttings, and solid granite tunnels eliminate line-of-sight to both terrestrial cellular base stations and orbiting satellites.

3. System Architecture & Ingestion Topology

RailConnect operates as a modular, containerized edge platform with microsecond IPC and real-time streaming:

Component	Technology	Function & Performance Role
Multi-WAN Edge Gateway	Python 3.12 / Linux Kernel	Interfaces with 4 physical modems, executes policy routing, handles packet buffering and failover transitions.

ML Inference Engine	Scikit-Learn / Random Forest	Runs at 1.0 Hz sampling rate; analyzes short-term trends to predict failure within a 10s lookahead window.
Train LAN QoS Orchestrator	IEC 61375 / Linux TC / FQ-CoDel	Shapes internal carriage consist bandwidth, shielding critical train signaling when cellular SIMs drop.
Time-Series Database	SQLite / WAL Mode	High-throughput rolling telemetry persistence for audit trails, compliance logs, and retrospective replay.
Real-Time API & Push	FastAPI / WebSockets	Broadcasts system snapshots to fleet control centers at 1.0 Hz with sub-5ms internal latency.

4. Heterogeneous Multi-WAN Uplink Matrix

RailConnect continuously arbitrates across 4 heterogeneous uplinks, avoiding vendor lock-in to any single carrier:

Uplink Identifier	Physical Medium	Frequency / Protocol	Throughput	Primary Operational Domain
5G Carrier Alpha	Cellular	5G C-Band (High Band)	Up to 75 Mbps	Dense urban departures, suburban mainline tracks.
5G Carrier Beta	Cellular	5G Sub-6 GHz (Coverage)	Up to 88 Mbps	Rural ridges, open valleys, secondary cell sectors.
Starlink LEO Rail-Link	Satellite	Ku/Ka Band (Low-Earth Orbit)	Up to 165 Mbps	Mountain canyons, rural open track without cell towers.
Trackside Wi-Fi 6E	Station Wi-Fi	6 GHz 802.11ax Gigabit	Up to 850 Mbps	Terminal stations and major depot platforms.

Composite Quality Score Formula

To evaluate health holistically, RailConnect computes a dynamic Composite Score (0–100) for every link:
Score = 0.35 × (Bandwidth / 150) + 0.30 × Latency_Score + 0.25 × Loss_Score + 0.10 × RF_Signal
Where *Latency_Score = max(0, 100 - (Latency - 5) × 0.45)* and *Loss_Score = max(0, 100 - Loss × 10)*. Links scoring below 30 or exhibiting packet loss > 90% are flagged as offline.

5. Dynamic Onboard Train LAN QoS Orchestration

A critical innovation in RailConnect is resolving the direct connection between fluctuating external SIM bandwidth and internal carriage LAN traffic. High-speed trains cannot guarantee uniform external throughput; hence, the Onboard Train LAN (IEC 61375 backbone) dynamically modulates its traffic slicing policy into three distinct modes:

LAN QoS Mode	External WAN Status	Critical Ops Allocation	Passenger Wi-Fi Policy	Carriage AP State
Unconstrained Gigabit	Station Wi-Fi or Strong 5G (> 60 Mbps)	4.2 Mbps (100% SLA)	Unthrottled HD streaming, high-speed passenger web access.	**Healthy (Green)**
Adaptive SIM Throttling	Cellular Doppler / handoff fade (25–60 Mbps)	3.5 Mbps (100% SLA)	Dynamic shaping: compresses video/media; preserves messaging/browsing.	**Throttled (Amber)**
Critical Failsafe	Tunnel or RF blackout (< 20 Mbps)	1.0 Mbps Guaranteed	Paused (0.0 Mbps) to strictly protect train signaling and driver radio.	**Failsafe (Red)**

The consist distribution spans 5 carriage segments: **Locomotive Cab** (ETCS & Driver Radio, 6 devices), **Coach C1** (Executive, 52 pax), **Coach C2 & C3** (Standard class, 172 pax total), and the **Pantry & Systems Car** (CCTV recorders, Passenger Information Screens). Each coach Access Point dynamically receives allocated bandwidth shares, ensuring fair distribution under all conditions.

6. Evaluator Simulation & Handover Lifecycle

To rigorously validate the system, an accelerated 16.0 km corridor simulation was executed across 75 seconds. The experimental timeline proves rapid autonomous switching between carriers and immediate LAN protection:

Time	Corridor Segment	Active Uplink	WAN Bandwidth	LAN QoS Mode	Pax LAN Throughput
0s – 7s	Metro Departure (km 0.0–1.5)	5G Carrier Alpha	72.0 ➔ 11.0 Mbps	Unconstrained ➔ Throttling	53.8 ➔ 18.2 Mbps
7s – 24s	Ridge Mainline (km 1.5–5.0)	5G Carrier Beta	88.0 Mbps (Stable)	Unconstrained	68.4 Mbps
24s – 38s	Canyon Pass (km 5.0–7.5)	Starlink LEO Rail-Link	165.0 Mbps	Unconstrained	135.2 Mbps
38s – 55s	Mount Kestrel Tunnel (km 7.5–11)	All Links Blackout	0.0 Mbps	**Critical Failsafe**	**0.0 Mbps (Paused)**
55s – 75s	Alpine Terminal (km 11–16)	Trackside Wi-Fi 6E	850.0 Mbps	Unconstrained Gigabit	830.5 Mbps

7. Collaborative Data Request & Pilot Evaluation Proposal

To benchmark RailConnect against live trackside environments, we are requesting operational modem telemetry datasets from railway administrations, transit authorities, and onboard router manufacturers (Peplink, Cradlepoint, Advantech, Cisco).

Target Telemetry Parameters Requested

Domain	Requested Field / Metric	Sampling Frequency & Importance
Cellular Modem Telemetry	RSRP (dBm), RSRQ (dB), SINR (dB), Ping Latency (ms), Jitter, Uplink/Downlink Mbps, Carrier PCI / Cell ID.	1.0 Hz to 0.2 Hz (every 1–5s). Essential for training Doppler attenuation curves.
GPS & Geolocation	Timestamp (UTC), Latitude, Longitude, Ground Speed (km/h), Heading, Track Milepost/Kilometer.	1.0 Hz. Correlates cellular fades with physical terrain, cuttings, and tunnels.
Router Transition Logs	SIM failover event logs, WAN switch timestamps, packet loss drops, WAN bonding status.	Event-driven. Used to benchmark reactive threshold algorithms vs ML proactive models.
Coach LAN Utilization	Connected Wi-Fi client counts per carriage, total aggregate throughput (Tx/Rx Mbps).	Periodic (5–30s). Validates carriage traffic shaping and QoS starvation rules.

Data Privacy, Security & Mutual Value Commitment

- **Zero Passenger PII:** RailConnect requires strictly physical modem RF layer metrics and network-layer stats. No passenger browsing history, MAC addresses, or personal identifiable information (PII) is ever collected or processed.
- **Confidentiality & Non-Disclosure:** All submitted data will be protected under mutual Non-Disclosure Agreements (NDA). Cell tower IDs and exact rail GPS coordinates can be pseudonymized or normalized upon request.
- **Deliverables Returned to Participating Organizations:** In return, our research group will provide an executive **Corridor RF Health & Predictive Handover Audit Report**, including dead-zone heatmaps, simulated passenger QoS improvements, and optimal router handover configurations for your specific routes.

RailConnect Project Collaboration Office
Inquiries: contact@railconnect.io • Repository: github.com/ashuyadav/railconnect-dashboard
System Demo Live Instance: http://localhost:8000 • Architecture: FastAPI / React / WebSockets / SQLite